BIOLOGY

Max.Marks:100 Time Allowed: 3 hrs

Theory: 70 Marks Practicals: 30 Marks

SECTION A: BOTANY Marks: 35

Unit-I. Diversity of Life 9 Marks

Biodiversity: Variety of living organisms; Need and History of classification – Artificial, Natural and Phylogenetic classifications. **Biosystematics**: Taxonomy and Systematics; Concept of species and Taxonomical hierarchy; Binomial nomenclature; Herbarium.

Two Kingdom and Five kingdom classifications and their merits and demerits; General characters and classification of Monera, Protista and Fungi; Lichens; Status of Viruses, and Viroids.

Unit -II. Kingdom Plantae 9 Marks

Classification of plants into major groups; General characters of Algae, Bryophyta, Pteridophyta and Gymnosperms and their classes.

Morphology of Flowering plants and their function: Morphology of root, stem and leaf (without their modifications); Morphology of Inflorescence, flower, fruit and seed. Description of family *Solanaceae*.

Unit - III: Plant Anatomy 5 Marks

Tissue systems in plants – Epidermal, ground and vascular tissue systems; Anatomy and functions of dicot and monocot root, stem and leaves.

Unit-IV Plant Physiology 12 Marks

Respiration: Cellular respiration; Glycolysis, Kreb's cycle and Electron transport system (along with ATP energetics); Chemiosmotic hypothesis; Aerobic and Anaerobic respirations; Amphibolic pathways; Respiratory quotient.

Photosynthesis: Historical background; Site of Photosynthesis; Photosynthetic pigments; Mechanism of Photosynthesis – Light dependent phase (Light reaction), Photosystems; Cyclic and non-cyclic photophosphorylations; Light independent (biosynthetic) phase –

Calvin (C_3) cycle and Hatch & Slack (C_4) cycle; Factors affecting photosynthesis; Photorespiration.

Plant Growth and Development: Characteristics of plant growth; phases of growth, growth rate, growth curve; conditions of growth; differentiation, dedifferentiation and redifferentiation.

Sequence of developmental process in a plant cell; Plant growth regulators; Discovery and physiological effects of Auxins, Gibberellins, Cytokinins, Ethylene and Abscissic Acid.

SECTION B: ZOOLOGY — Marks: 35

UNIT I: Diversity in Living World — 8 Marks

Animal Kingdom; General characters and classification of animals (non-chordates up to phyla level and chordates up to the level of class).

National Parks with special reference to Dachigam, Kishtwar, Salim Ali, Kazinag and Hemis.

Concept of Sanctuaries and Biosphere reserves.

Unit - II: Structural Organisation and Animal Biomolecules — 7 Marks

Morphology, Anatomy and functions of digestive, circulatory, respiratory, nervous and reproductive systems of Frog.

Basic chemical constituents of living bodies; Bio-molecules: Structure and function of Carbohydrates, proteins, lipids and Nucleic acids; Primary and secondary metabolites.

Enzymes: Types, Properties and functions

Unit-III: Cell Structure and Function — 8 Marks

Cell: Brief description of cell; Cell theory; Prokaryotic and Eukaryotic cell, Cell wall, Membrane and Cell organelles (Plastids, Mitochondria, E.R., Golgi, Ribosomes, Lysosomes, Nucleus, Vacuoles, centrioles and Cytoskeleton); Cilia and Flagella; Nuclear organisation.

Cell Division: Cell Cycle, Mitosis and Meiosis.

Unit-V Human Physiology — 12 Marks

Breathing and Respiration: Respiratory system in humans; mechanism of breathing and its regulation in humans; respiratory volume; Exchange of gases and their transport; Respiratory disorders.

Body fluids and Circulation: Composition of blood; Blood groups and Rh factor; Lymph; Human Circulatory system; Cardiac cycle and ECG; Double circulation; regulation of cardiac activity; Disorders of circulatory system.

Excretory Products and their Elimination: Modes of excretion; Human excretory system; Urine formation, osmo regulation; regulation of kidney function; Urinary disorders; and artificial kidney.

Locomotion and Movement: Types of movement; Muscle, Contractile proteins and Muscle contraction (Mechanism); Skeletal system and its functions; Joints; Disorders of muscular and skeletal systems.

Neural Control and Coordination: Neuron and nerves; Nervous system in humans; Nerve impulse.

Chemical Coordination and Integration: Human endocrine system; Hormones of Heart, Kidney and G.I. Tract; Mechanism of hormone action (elementary idea); Hormonal disorders.

*(**Note**: diseases of human physiology systems to be taught in brief)*

Scheme of Assessment (Theory)

BOTANY

Question paper contains four sections namely

1. **Section –A** contains 5 (Objective Type Question/Multiple Choice Questions) of 1 mark each. 1x 5= 5 marks
2. **Section- B** contains 5 Very Short Answer Type Questions of 2 marks each to be answered in20 to 30 words. 2 x 5 = 10 marks
3. **Section – C** contains 5 Short Answer type questions of 3 marks each to be answered in 100 to 150 words. 3 x 5 = 15 marks
4. **Section - D** contains 1 Long Answer type question of 5 marks each to be answered in 150to 200 words. (With Internal Choice) 5 x 1 = 5 marks

Note: The paper setter shall incorporate Higher Order Thinking Skills (HOTS) questions apart from knowledge, memory and application based questions.

ZOOLOGY

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PRACTICALS **Maximum Marks: 30**
Time: 03 Hours **External: 20 Marks/ Internal: 10 Marks**

SECTION A: BOTANY

Max. Marks: 15

A. Observation/ Spotting

1. Study of different parts of a Compound Microscope.
2. Study of Specimens and identification with reasons – Bacteria, Algae (*Spirogyra*), Fungi (*Rhizopus*, Mushroom & Yeast); Lichens; Bryophytes (*Marchantia* & *Funaria*); Pteridophyta (*Dryopteris*); Gymnosperms (*Pinus* – male & female cones); Angiosperms (one monocot & one dicot plant).
3. Study of T.S. of dicot and Monocot Root, Stem and Leaf from permanent slides.
4. Study of Leaf (arrangement, shape & venation) and Leaf types (simple & compound).
5. Study and identification of types of infloresences (Cymose & Racemose).

B. List of Experiments:

1. Description of locally available flowers from the families Solanaceae and Liliaceae.
2. Study of Osmosis by Potato Osmoscope (Osmometer).
3. Study of plasmolysis in epidermal peels (e.g. Onion/Rhoeo/lily leaves).
4. Study of imbibition in seeds/raisins.
5. Study of distribution of stomata on the upper and lower surfaces of leaves.
6. Separation of plant pigments through paper chromatography.
7. Study of the rate of respiration in germinating seeds.

C. Project work

1. Collection and preservation of plant specimens for Herbarium.
2. Biodiversity trip: Visit to a botanical garden/flora rich area and preparation of report.

3. Observations on the experimental set up on Phototropism, Apical bud removal, Anaerobic respiration and Suction due to transpiration.

SECTION – B: ZOOLOGY

Max. Marks: 15

A. Observation/ Spotting

1. Study and handling of Compound Microscope.
2. Study of specimens/ Virtual specimens/ Models and identification with reasons – *Amoeba*, *Paramecium*, Hydra, Liver fluke, *Ascaris*, Leech, Earthworm, Honey bee, Snail, Starfish, Shark, *Labeo*, Frog, Lizard, Pigeon and Rabbit.
3. Study of Animal cell and its organelles with the help of Slides/ Charts.
4. Study of Mitosis and Meiosis from prepared slides.
5. Study of organ systems of Frog with the help of Charts/ Models/Virtual Dissection.
6. Study of Human skeleton and different types of joints (virtual images/models only).

B. Experiments

1. To Test for presence of urea in urine.
2. To test the presence of carbohydrates and protein (albumin) in Urine sample.
3. Preparation and study of human blood smear.
4. To test the presence of sugar in urine/blood sample.

C. Project work:

4. Biodiversity trip: Visit to a zoological garden/ National Park and preparation of report.
5. Study of cyclosis in Paramecium.

BOOK PRESCRIBED:

1. *A textbook of Biology for class XI published by NCERT, New Delhi.*